Josh Talks Growth Assignment - Workspace Audit

Technical Integrity, Evidence Triangulation, and Scope Boundaries

1. Primary Data Sources & Scope Disclosures

- **Supplied Workbook:** Deepgram_Failure_Analysis_Data.xlsx contains 7 sheets (Read me, Sessions, Results, Transcripts, Deepgram API check, Audio quality, Deepgram settings) joining to 15 recording IDs.
- **RTTM Zip Package:** Contains 60 files across 3 folders (reference, deepgram_nova3, silero_vad) covering 20 recording IDs. Five IDs are outside the workbook scope and excluded from workbook statistical aggregates.
- **Missing Source Assessment PDF:** The original assessment PDF referenced in instructions is not present in the workspace, representing an audited workspace limitation.

2. Quality Control & File Integrity Checks

- **RTTM Timing Validity:** 100% of the 60 RTTM files contain valid non-negative timestamps and positive segment durations. No timing corruption detected.
- **Empty Output Files:** Exactly two Nova-3 RTTM files are empty (0 bytes, 0 segments): D20260825_S003_015 (workbook ID) and D20260825_S003_012 (package-only ID).
- **Component Identity Verification:** Workbook DER matches (Miss % + FA % + Conf %) within 0.01 percentage points across all 15 rows due to displayed rounding.
- **API Check Findings:** API check sheet reports zero warnings for inspected rows. However, this only logs provider error strings; it does not prove upstream acoustic integrity.

3. Reproducibility & Pipeline Automation

- Re-running Supporting_Work/analysis/analyze_dataset.py deterministically regenerates consolidated tables, correlation matrices, and RTTM validation files.
- Re-running Supporting_Work/analysis/independent_quality_audit.py verifies mean DER (64.4506...%), median DER (86.02%), and worst recording subsets.
- Re-running generate_deliverables.py compiles all submission-ready PDFs, Word documents, mockups, and execution guides.